CIS 121 Assignment – Reading Data from a File. Practice Pass by Value Functions. For each problem develop an IPO within this document. Save the document and upload to Blackboard for grading. Then develop a C++ program for each problem and upload the .cpp file to Blackboard.

1. Read in auto make, model, MSRP (manufacturer suggested resale price) and sales price from a file. See auto.txt below. For each auto (line of data) compute the savings to be MSRP – sales price and sales tax in a single function.

Pass to the function MSRP, sales price by value, sales price and sales tax by reference. The function will:

1. Compute savings = MSRP – Sales Price
2. Compute sales tax = Sales Price * 0.07

For each auto (line of data) display make, model, MSRP, sales price, savings and sales tax. Sum the savings.

After all autos (lines of data processed) display sum of savings.

Auto.txt

Honda Accord 25000.00 22000.00

Honda CRV 30000.00 28000.00

Toyota Corolla 28000.00 25000.00

Ford Fusion 23000.00 20000.00

Input	Process	Output
Msrp	Tax_rate	

Sales price
Car
Model
	Void calcsavings(double msrp, double salesPrice, double &salesTax, double &salesPrice)
	{
	Savings = msrp – salesPrice
	SalesTax = salesPrice * tax_rate
	
	}
		Display table

Savings
Sales tax
	While (inFile >> car >> msrp >> salesPrice)
	
	{
	Sum_savings+=savings
	SumSalesTax += salesTax
	
	Recall cout <<
	}
	InFile.close();
		Sum of total savings:

Sum savings Sum sales tax		

Read in the grocery item, quantity and cost per item from a file. See grocery.txt below. For each item determine the extended price (quantity * cost per item) within a function.

Pass to the function quantity and price. The function will:

3. Compute extended price = quantity * price
4. Return extended price

For each item (line of data) display item, quantity, cost per item and extended price.

Also, sum the extended price.

After all lines of data are processed, compute tax of 7% (sum of extended price * 0.07). Compute total receipt to be sum of extended price + tax. Display sum of extended price, tax and total receipt.

Grocery.txt

Banana 10 0.50

Oranges 5 0.10

Cookies 15 0.30

Hamburgers 4 1.00

Buns 4 0.50

Soda 10 0.99

Input	Process	Output
item	Const double tax_rate = 0.07	Table Grocery Item Quantity Cost Per Item Extended Price

Qty	Double calcExt (double qty, double price
	{
	Return Ext_price
	}
		Sum of extended price:
		Tax:
		Total Receipt:

price	While (inFile >> item >> qty >> price) Double ext_price = recall function SumExtPrice+=ext_price Recall cout << } InFile.close();	
Extended price		
Sumextprice = 0.0		

Tax amnt		

3. Read gallons of gas used and miles travelled from a file. See trips.txt below. For each trip compute miles per gallon within a function.

Pass to the function gallons and miles. Within the function:

1. Compute mpg = miles / gallons.
2. Return mpg.

For each trip (line of data) display gallons, miles and mpg.

Sum miles travelled and gallons of gas used.

After all data is processed (end of file) display sum of gallons and sum of miles travelled.

Trips.txt

20 500

10 600

5 50

10 350

Input	Process	Output

Gas used	Double calcMPG(double gas, double miles)
	{
	Return Miles/gallons
	}
		Table:
		Gallons
		
		Miles
		
		Mpg

Miles travelled	Ifstream inFile (trips.txt)	

Miles per gallon	While (inFile >> miles >> gallons)
	{
	Recall function (create data type for mpg)
	
	Total_miles+= miles;
	Total_gallons+= gallons;
	Cout <<
	
	}
	InFile.close();
		Sum of Gallons used:
		
		Sum of Miles travelled:

Total_miles= 0.0		
Total_gallons = 0.0		

4.Read last name and annual salary from a file. See empl.txt below. For each employee (line of data) compute their bi-weekly pay within a function.

Pass to the function the annual salary. The function will:

1. Compute bi-weekly salary = annual salary / 26
2. Return bi-weekly salary

For each line display last name, annual salary and bi-weekly salary. Sum the annual salaries. Count the number of employees (each line of data).

After all employees processed compute average annual salary (total annual salary / count of employees). Display sum of annual salary, count of employees and average annual salary.

Empl.txt

Jones 50000.00

Adams 65000.00

Baker 45000.00

Smith 75000.00

Input	Process	Output
Lname Annual salary	Const int pay periods – 26	Last name Annual salary Bi weekly salary
Bi-weekly	Double calcBiweekly (double annualSalary) Annual / pay period	- Close loop -
Salary sum	Ifstream inFile(emply.txt)	Average annual salary Annual salary sum Employee entry count

Count	While (inFile >> lname >> annual salary)
	{
	CalcBiweekly function
	And
	Summation code
	}	

	Double averagesalary(count >0)	

5.He input to this program is last name, student code and credits taken. See file student.txt below.

You are to read each line of data into your program (one line at a time of course). Write a function to calculate tuition owed and course fees in a single function. You must pass to the function student code and credits taken by value and tuition owed and course fees by reference.

Within the function compute tuition owed and course fees:

1. Determine cost per credit hour. In district students (student code == 'I') pay $250.00 per credit hour. Out of district students (student code == 'O') pay $500.00 per credit hour.
2. Calculate tuition owed to be credit hours * cost per credit hour
3. Compute course fees as 10% of the total tuition.

Also, within the program sum the tuition owed for all students. Count the number of students. For each student (line of data), display last name, student code, cost per credit hour, credits taken, tuition owed and course fees. Additionally, sum the tuition owed.

After all students are processed (end of loop), calculate the average tuition cost per student (sum of tuition owed / number of students). Display sum of tuition owed, number of students and average tuition cost per student.

Student.txt

Jones I 15.00

Baker O 20.00

Davis I 10.00

Michaels O 12.00

Baez I 12.00

Input	Process	Output
Last name	Display constant value variables	

Student code	Void calcTuition (char student code, int credits, double &tuitionowed, double &coursefees)
	{
	Double costpercredit
	(student code == 'I')
	? In state tutiton
	: Out of state tuition
	
	TuitionOwed = credits * costper credit hour
	
	Course fees = tuitionOwed * fee percent
	}
		Last Name:
		Student Code:
		Cost/Credit:
		Credits Taken:
		Tuition Total:
		Course Fees:
		Tuition Sum:
		Number of Entries:

	ifstream inFile(student.txt)	
Credits	While (infile >> last name >> student code >> credits)	Average cost of Tuition per Student:
Tuition owed Course fees		
Tuition sum		
Count		